

# AfyPlus Generative Layer - Engineering Decision Memorandum

To: CTO and Medical Director

From: Platform Engineering (observability capstone)

Re: Six weeks of production AI without automated evaluation, drift monitoring, or cost tracking

Evaluation run mode: offline

## Executive summary

The AfyaPlus generative layer has been answering clinical and insurance queries without automated quality gates, drift reports, or token accounting. We evaluated both gpt-4o-mini and gpt-4o on a 15-question clinical set (30 model rows), simulated three months of traffic, and projected 30-day spend at 2,000 requests/day. Mean judge overall is 4.23/5 for mini and 4.81/5 for gpt-4o; Month 3 mean ROUGE-L is 0.3389 versus 0.4064 in Month 1; 30-day generative spend is \$25.22 against a \$1500 cap.

## Quality performance breakdown

Automated overlap (BLEU / ROUGE-L / token F1) plus an LLM judge (correctness, groundedness, relevance, helpfulness) were scored for every row. gpt-4o-mini passed 2 of 3 feature gates; gpt-4o passed 3 of 3. Mean groundedness is 4.47 (mini) vs 5.00 (gpt-4o). Clinically this means mini is cheaper on overlap-heavy protocol restatements but is more likely to drop numbers, waiting periods, or red-flag lists - the details clinicians need for safe routing.

## Cost and efficiency analysis

At the documented 75/25 canary, 30-day token spend is \$25.22 (\$0.84/day vs a \$50 daily cap). Routing each feature to the cheapest model that still PASSES the quality gate saves \$58.96 versus sending all traffic to gpt-4o and \$-5.91 versus the current canary. Human review of the same 59,940 tickets at an assumed \$2.50/ticket would cost \$149,850 - token inference remains far cheaper than full human validation, but it does not replace a clinician on failed gates.

Recommended routing from this run: chronic\_care\_referral -> gpt-4o; insurance\_routing -> gpt-4o-mini; pediatric\_triage -> gpt-4o-mini.

## Observability metrics (Prometheus)

The console publishes these gauges at GET /metrics from the same Phase 1-3 artifacts as this memo. Page WARN at 80% of cap and CRIT at 95%. afyaplus\_exceptions\_total is a process counter that starts at 0 when the console boots; this snapshot is 0 because artifact generation succeeded.

|                                                                               |               |
|-------------------------------------------------------------------------------|---------------|
| <b>afyaplus_health_up</b>                                                     | <b>1</b>      |
| 1 if evaluation artifacts are readable                                        |               |
| <b>afyaplus_exceptions_total</b>                                              | <b>0</b>      |
| Artifact load exceptions since boot                                           |               |
| <b>afyaplus_daily_spend_usd</b>                                               | <b>0.84</b>   |
| Mean daily generative spend                                                   |               |
| <b>afyaplus_monthly_spend_usd</b>                                             | <b>25.22</b>  |
| 30-day generative spend                                                       |               |
| <b>afyaplus_daily_cap_usd</b>                                                 | <b>50</b>     |
| Daily spend cap                                                               |               |
| <b>afyaplus_monthly_cap_usd</b>                                               | <b>1500</b>   |
| Monthly spend cap                                                             |               |
| <b>afyaplus_month3_rouge_l</b>                                                | <b>0.3389</b> |
| Month 3 mean ROUGE-L                                                          |               |
| <b>afyaplus_drift_flag{column="rouge_l"}</b>                                  | <b>1</b>      |
| 1 if ROUGE-L drifted in latest month                                          |               |
| <b>afyaplus_judge_overall{model="gpt-4o",feature="chronic_care_referral"}</b> | <b>4.80</b>   |
| Mean LLM-judge overall                                                        |               |
| <b>afyaplus_judge_overall{model="gpt-4o",feature="insurance_routing"}</b>     | <b>4.84</b>   |
| Mean LLM-judge overall                                                        |               |

|                                                                                    |             |
|------------------------------------------------------------------------------------|-------------|
| <b>afyaplus_judge_overall{model="gpt-4o",feature="pediatric_triage"}</b>           | <b>4.80</b> |
| Mean LLM-judge overall                                                             |             |
| <b>afyaplus_judge_overall{model="gpt-4o-mini",feature="chronic_care_referral"}</b> | <b>3.98</b> |
| Mean LLM-judge overall                                                             |             |
| <b>afyaplus_judge_overall{model="gpt-4o-mini",feature="insurance_routing"}</b>     | <b>4.24</b> |
| Mean LLM-judge overall                                                             |             |
| <b>afyaplus_judge_overall{model="gpt-4o-mini",feature="pediatric_triage"}</b>      | <b>4.48</b> |
| Mean LLM-judge overall                                                             |             |

## Systemic operational risks

First detected drift: Month 1, column input\_length. Month 3 latency mean is 1101.6 ms with ROUGE-L 0.3389. That pattern matches longer, more complex production questions than the pre-deploy baseline. Hallucination risk is highest where groundedness fails the 4.0 gate: unsupported dosages or cover terms must not ship. Waiting-period emergencies remain a policy exception (bypass cover checks) and must stay out of cost-only routing.

## Actionable engineering roadmap

1. Route by feature, not by default canary. Keep any FAIL feature on gpt-4o. Estimated 30-day saving vs all-gpt-4o: \$58.96, with groundedness held at the 4.0 safety bar.
2. Page on first-column drift. Alert already isolates Month 1 / input\_length. Wire the Prometheus gauges catalogued above to the 80%/95% budget WARN/CRIT policy so spend and quality share one on-call path.
3. Do not drop human review on pediatric\_triage FAIL rows. Token cost of the full 30-day corpus is \$25.22 versus \$149,850 for blanket human review; spend the human budget only on gate failures and Month 3 drifted slices.

\*Coursework snapshot - not a live AfyaPlus production certification.\*